

Secure Handhelds on Assured Resilient networks at the tactical Edge (SHARE) Mission-Integrated Network Control (MINC)

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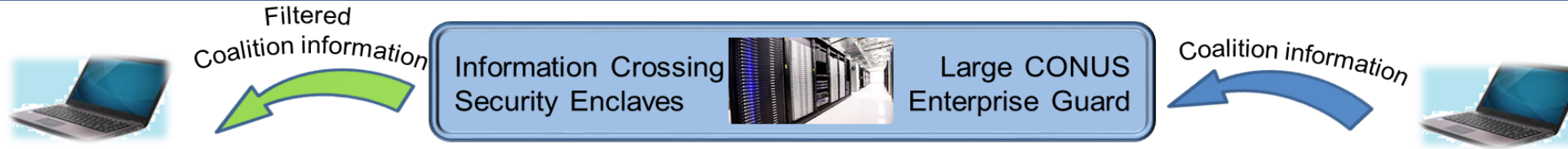
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DARPA SHARE Overview

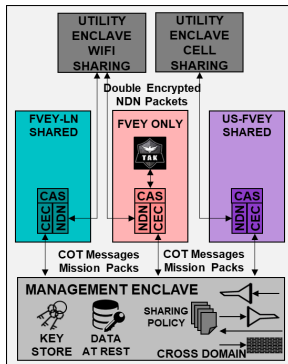
SHARE Overview

Today

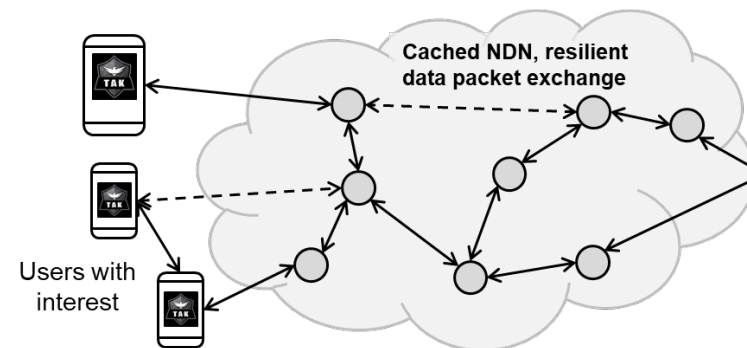


- Centralized, enterprise level approach has no way to establish trust at the tactical edge
- VPNs used to secure networks and devices quickly fail in high error rate tactical networks
- Configuring and deploying networks for US and coalition use takes months to years

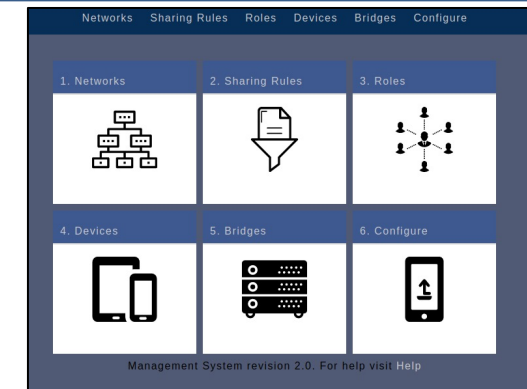
SHARE



Multiple levels of security on a single, trusted handheld



Resilient networking with built-in encryption, caching, and key distribution

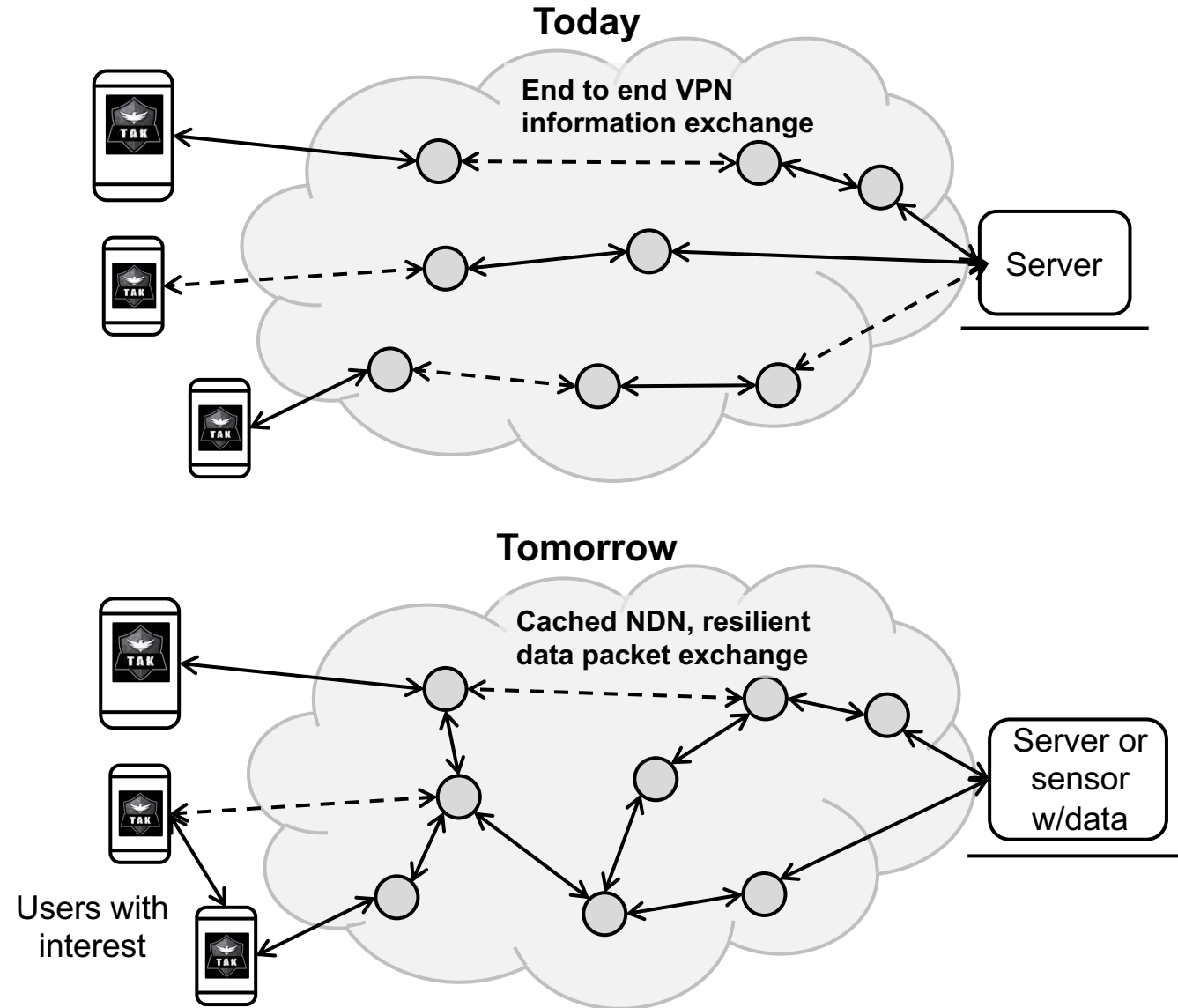


Software to automate and drastically reduce configuration time

Secure data sharing at multiple security levels between coalitions across mobile tactical networks

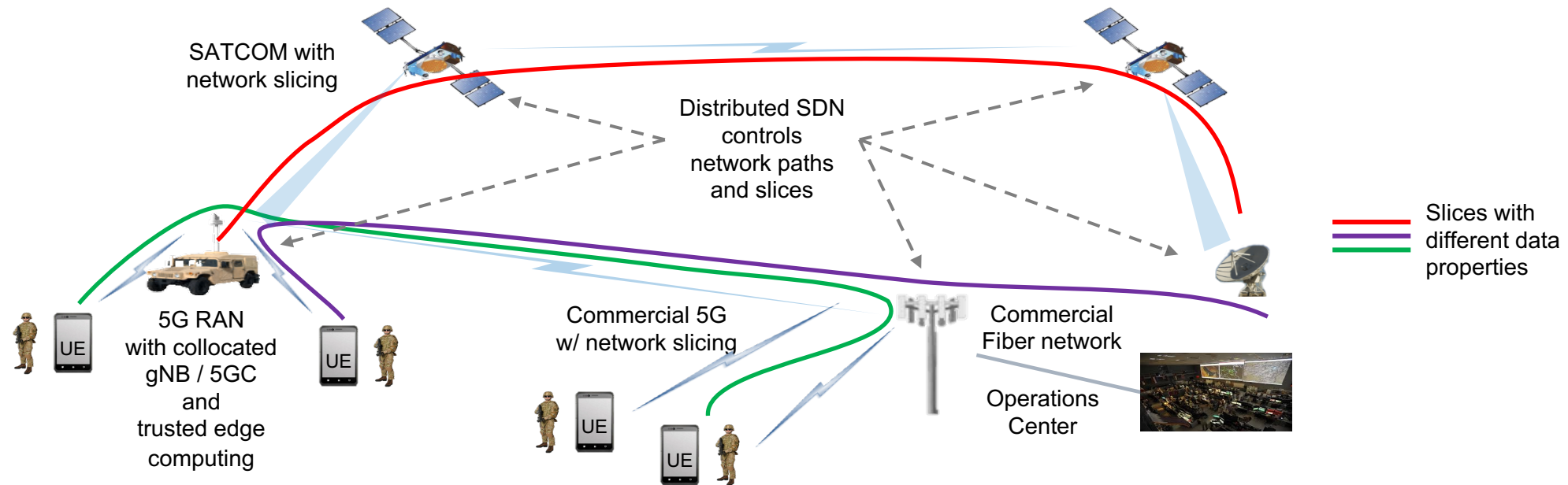
- Primary features of tactically hardened and secured Named Data networking (NDN) include:
 - Resilient and efficient packet forwarding to address mobility, i.e. links may be unstable but data packets, thus information flow, persist in the network
 - Rapid information recovery via data synchronization tailored to tactical network traffic patterns
 - Dual layer packet level security implementation that aligns with IPsec VPN data protections
 - Secure bridging of SHARE and non-SHARE networks and devices via NDN

Provides **security at the packet level** as each is authenticated, encrypted, and can securely transit the network on its own without a persistent, end-to-end connection



5G Secure Networking with SHARE

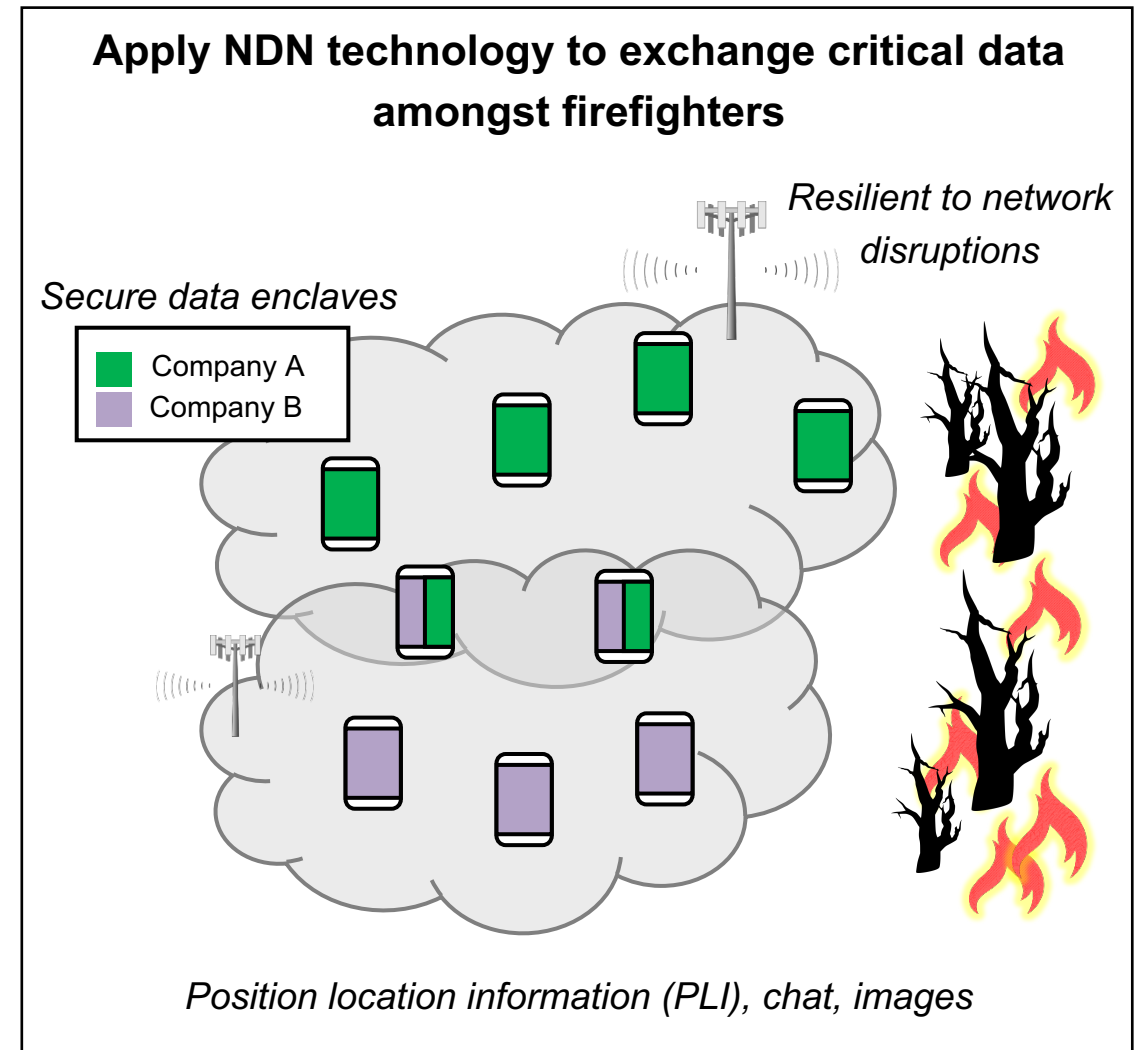
- Joint effort between OUSD(R&E), DARPA, and NSF
- 5G cellular network technology poised to support critical military applications
- Goal: Demonstrate the secure exchange of information over 5G commercial and military networks employing network slicing to further isolate data



SHARE Transition



- Promoting SHARE technology within user communities to demonstrate solutions to requirements, interoperability, and drive adoption
 - Actively engaged with the 75th Ranger Regiment and the Telemedicine and Advanced Technology Research Center on transition efforts
- DARPA working with NIST to ensure ease of access and adoption of SHARE NDN
 - Investigating collaboration opportunities related to public safety use case(s)
 - Demo SHARE NDN with TAK for shared situational awareness in disconnected networks



DARPA MINC Overview

MINC Problem Statement



Static networks limit mission execution

- Need to **dynamically control** networks of networks on-demand

Manual network configuration does not scale and is prone to error

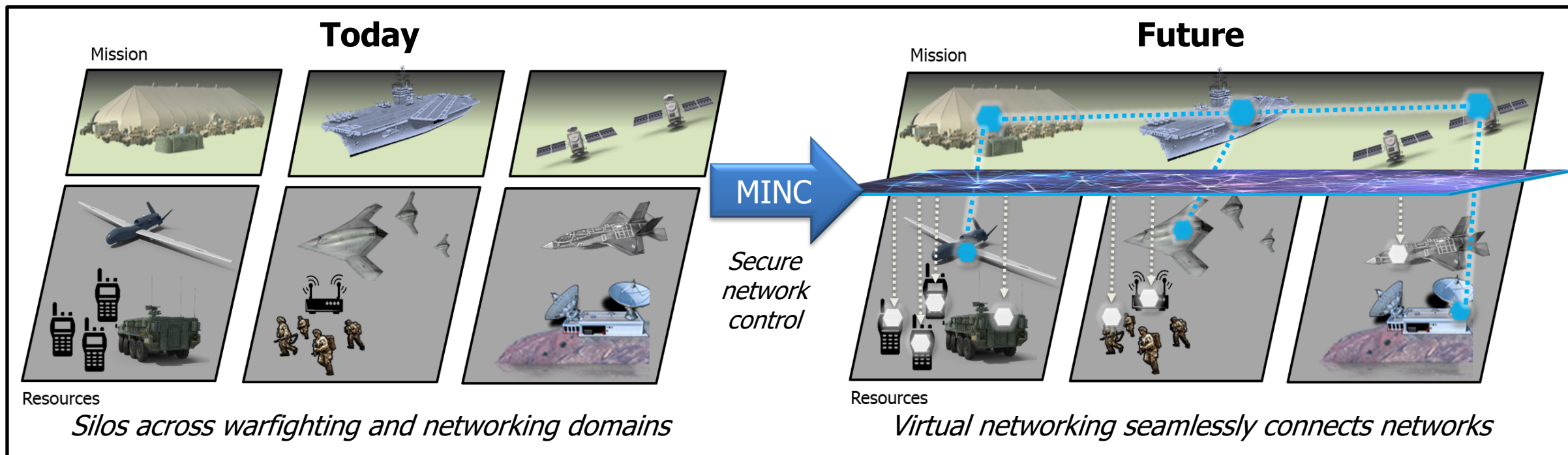
- Need to **manage complexity**



Cross-domain kill webs are only as effective as the communications enabling the warfighting services

MINC will ensure that critical data finds a path to the right user at the right time

Future Network Control



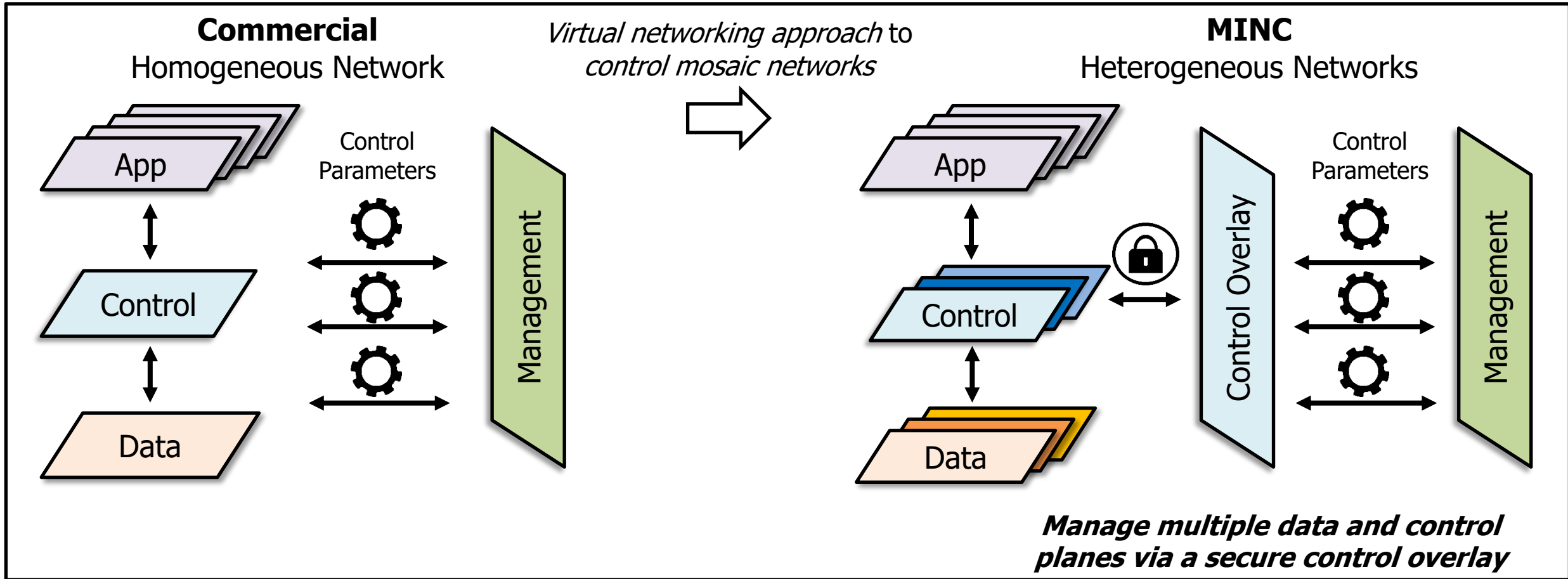
MINC will enable autonomous network discovery and mission-driven control

New mission-driven approach for managing network configuration and information flows

- “Always on” network overlay to achieve backward and forward compatibility across heterogeneous networks
- Connectivity realized “on-demand” between kill web services

A New Approach to Network Control

Software-Defined Networking (SDN) enables remote, intelligent control of heterogeneous networks

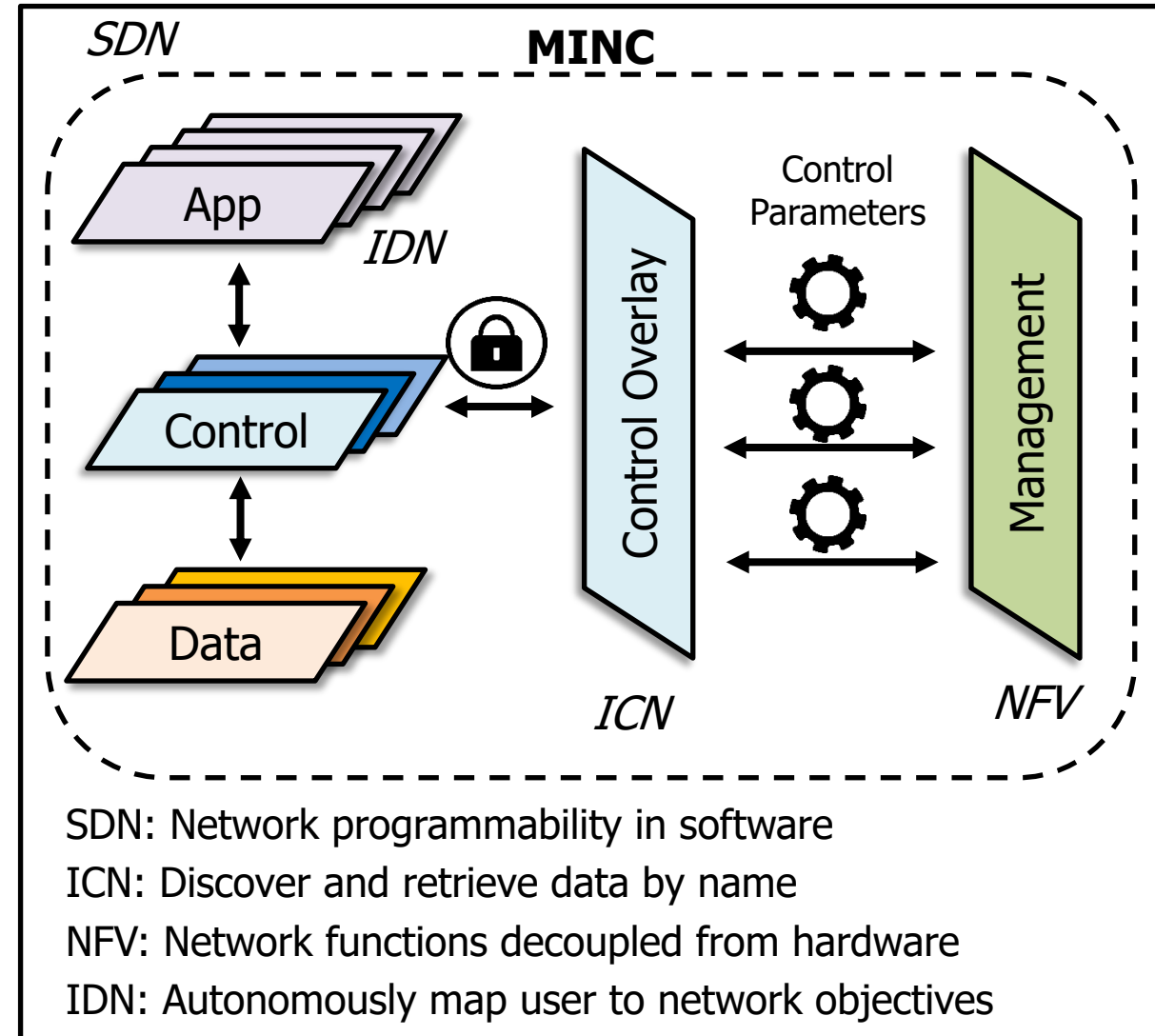


SDN construct presents a new opportunity to dynamically control and manage networks

MINC Technology Enablers

- SDN construct with an “always on” network control overlay to enable **adaptive control of networks**
- Information-Centric Networking (ICN) for **resilient delivery** of network control data
- Network Function Virtualization (NFV) to deploy virtual edge functions to **overcome insufficient network capacity**
- Intent-Driven Networking (IDN) to adapt network and applications according to **mission objectives**

Commercial investment is driving intelligent software-enabled networking albeit for less dynamic, *homogeneous networks*



MINC Focus Areas

Key Challenges:

Lack of network interoperability across heterogeneous communications systems at scale

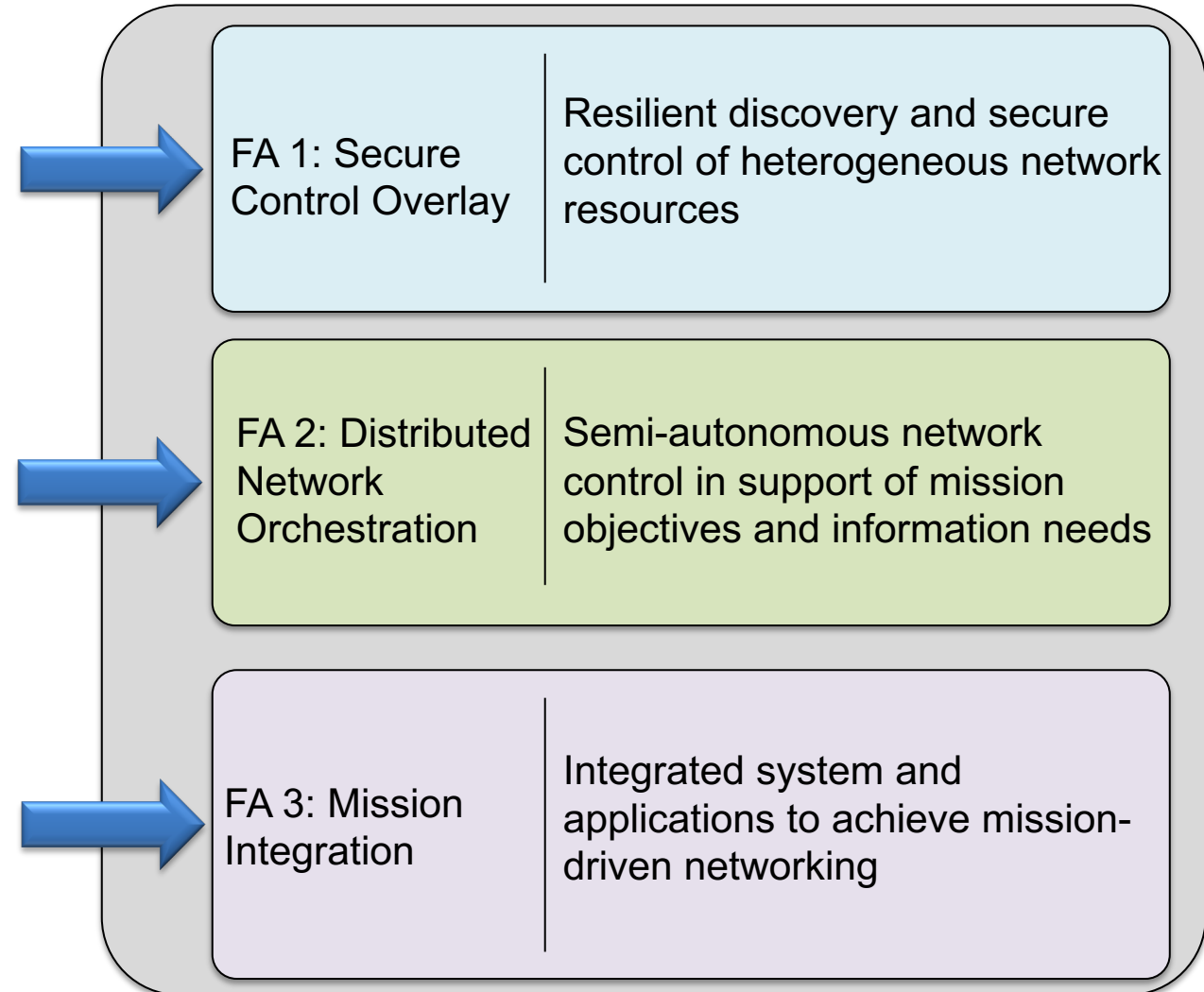
Insufficient network capacity to support missions

Adapting to Mission Dynamics

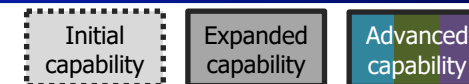
Key Insights:

- Software programmability
- Secure, Resilient ICN
- Control parameters across OSI layers including physical/virtual movement
- Jointly manage network and data
- Commercial IDN for military missions
- Adapt applications and data flows

Focus Areas:

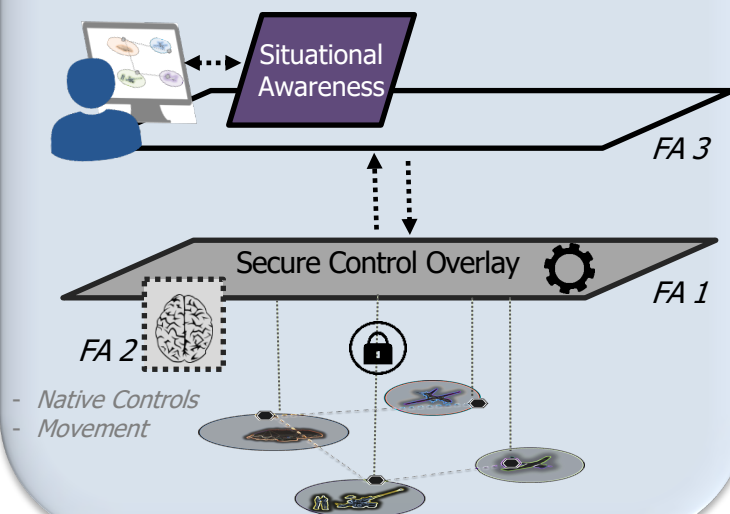


MINC Development Concept



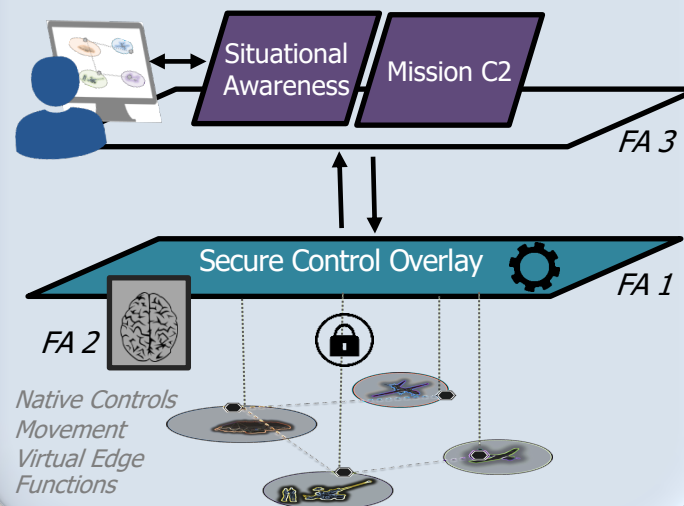
Phase 1 *Minimum Viable Product* Laboratory Demo

- Secure control framework
- Stand-alone network orchestration
- Blue Force SA Application



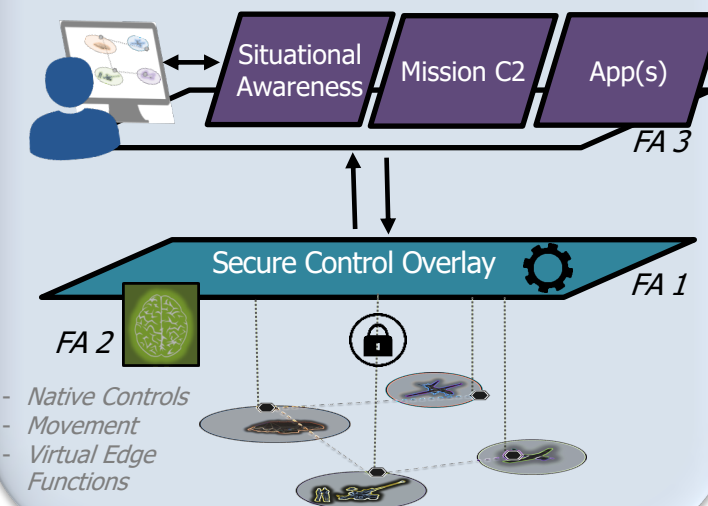
Phase 2 *Integrated System* LVC Demo

- Secure control in a highly dynamic environment
- Integrated network orchestration
- Integrated Mission C2 Application



Phase 3 *Transition-Ready System* Field Demo

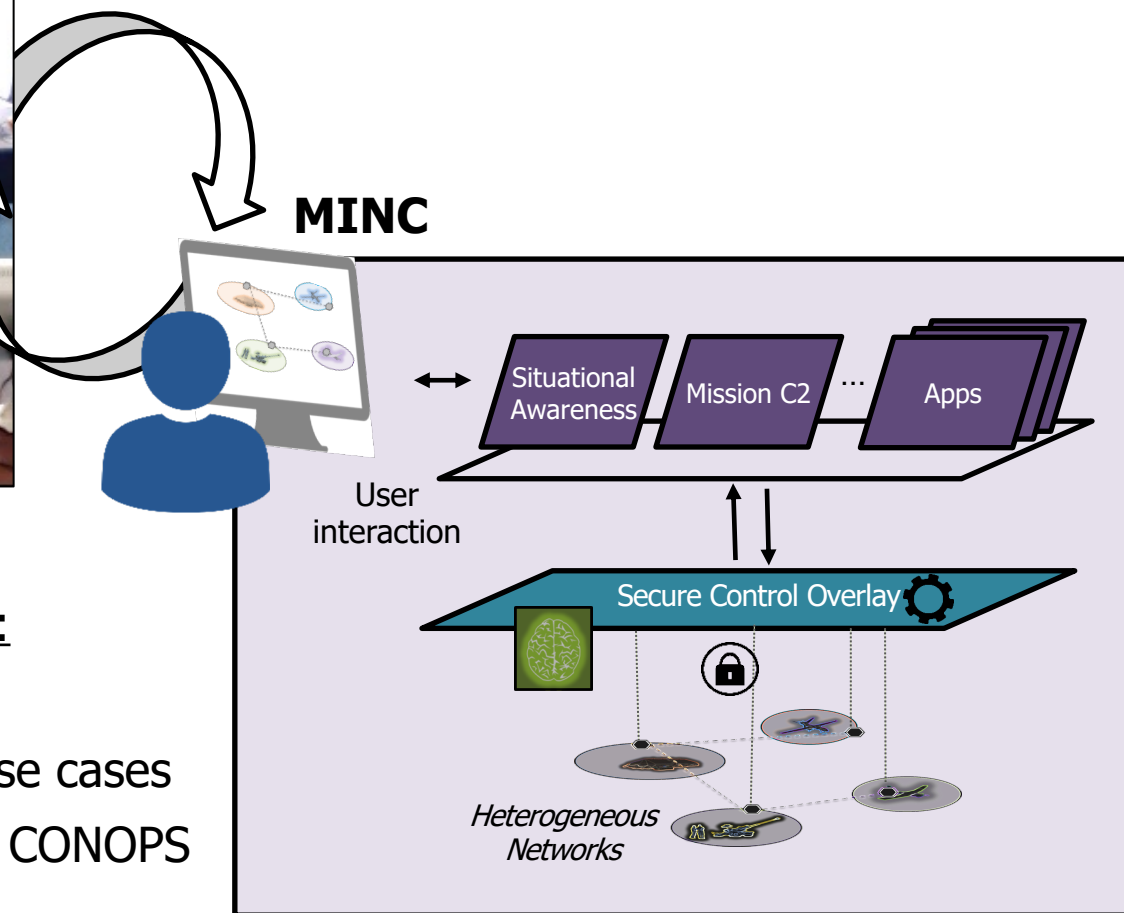
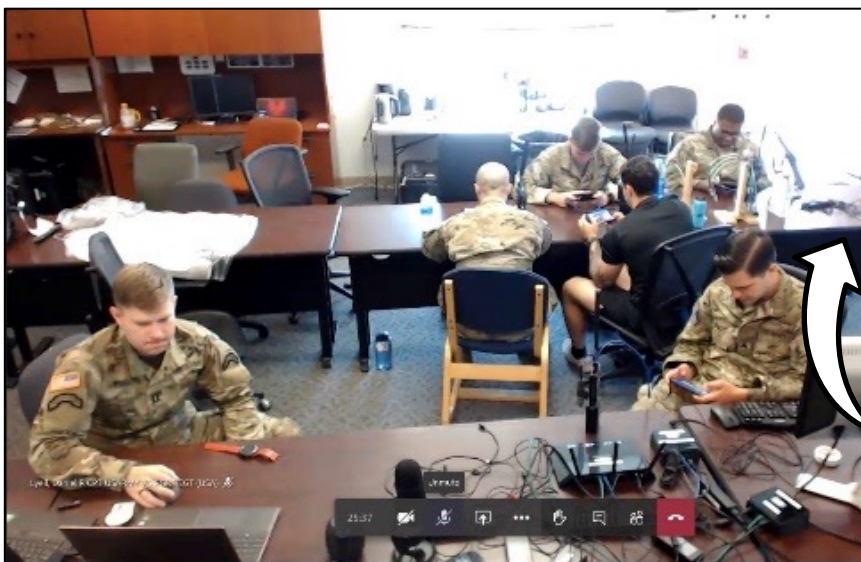
- Secure control in a highly dynamic, highly contested environment
- Advanced network orchestration
- Transition Application(s)



Development Security Operations (DevSecOps)



DevSecOps model drives frequent interaction and empowers transition partners



Create program mechanisms and behaviors that:

- Influence MINC features and applications
- Informs concept of operations (CONOPS) & use cases
- Creates real-time adaptability to support new CONOPS



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